

# Fractals

Infinite Details from Simple Rules

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CSE 200 Presentation

# History

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# A Crisis in 19th-Century Mathematics

For centuries, calculus rested on an assumption that felt obvious:

## The Assumption

Any continuous curve is **smooth almost everywhere** — zoom in far enough on any point, and the curve should straighten out into a well-defined tangent line.

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In 1872, **Karl Weierstrass** broke this assumption — on purpose.

He constructed a function that is **continuous everywhere**, yet **differentiable nowhere**: infinitely jagged, at every single scale, with no tangent line anywhere at all.

# Mathematical "Monsters"

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For nearly a century, these were considered **beautiful but useless** — curiosities built to stress-test the foundations of mathematics, not to describe anything real.

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Mandelbrot's insight: this isn't a flaw in the coastline — it's a sign that classical length and area are **the wrong tools** for something like this. What a coastline actually needs is exactly the "useless" mathematics of Cantor, Koch, and Hausdorff.

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- He had a real problem no one had connected to these ideas.
- He had **computers** — something Cantor, Koch, and Julia never had — letting him *see* these shapes for the first time.
- He united a century of scattered "monsters" under one name: **fractal**, from the Latin *fractus* ("broken").

# Definition

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# What are Fractals?

## Definition

A **fractal** is a geometric shape containing detailed structure at arbitrarily small scales, usually having a **fractal dimension** strictly exceeding the **topological dimension**.

## Key Characteristics:

- **Scale invariance**: Details at every level of magnification
- **Fractal dimension**: More than the *conventional* dimension
- **Infinite complexity**: Generated by simple, recursive rules
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*"Clouds are not spheres, mountains are not  
cones, coastlines are not circles..."*

— **Benoit Mandelbrot**

## Some Fractals

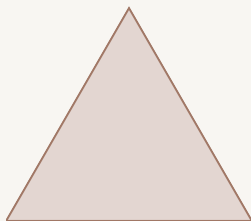
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# The Sierpinski Triangle

Let's construct the Sierpinski Triangle:

## Construction Process

1. Start with an equilateral triangle



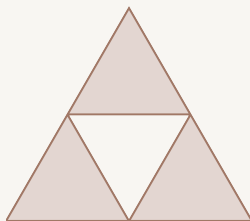
Iteration 0

# The Sierpinski Triangle

Let's construct the Sierpinski Triangle:

## Construction Process

1. Start with an equilateral triangle
2. Remove the central triangle (triangle connecting midpoints of the sides)



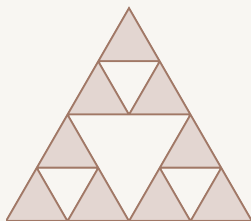
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## Construction Process

1. Start with an equilateral triangle
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3. Repeat for each triangle



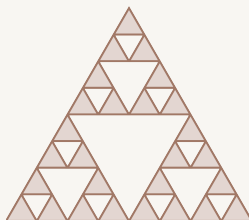
Iteration 2

# The Sierpinski Triangle

Let's construct the Sierpinski Triangle:

## Construction Process

1. Start with an equilateral triangle
2. Remove the central triangle (triangle connecting midpoints of the sides)
3. Repeat for each triangle
4. After infinite iterations, we get the Sierpinski Triangle



Iteration 3

# **Area & Perimeter**

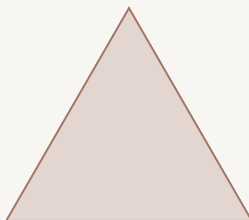
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# The Sierpinski Triangle: Area

Now let's find the area of triangle(s):

## Area Calculation

$$A_0 = \frac{1}{2} \cdot \sin(60^\circ) \cdot a^2 = \frac{\sqrt{3}}{4} a^2$$



Iteration 0

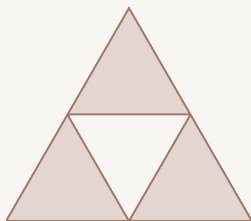
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$$\begin{aligned} A_1 &= A_0 - \frac{1}{2} \cdot \sin(60^\circ) \cdot \left(\frac{a}{2}\right)^2 \\ &= A_0 - \frac{\sqrt{3}}{16} a^2 = A_0 - \frac{1}{4} A_0 \\ &= \frac{3}{4} A_0 \end{aligned}$$



Iteration 1

We cut out one fourth of the area at each step.  
So, naturally area is reduced by  $\frac{3}{4}$  at each step.

# The Sierpinski Triangle: Area

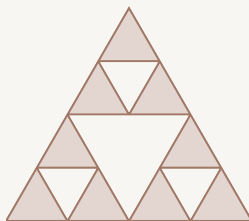
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## Area Calculation

$$A_0 = \frac{1}{2} \cdot \sin(60^\circ) \cdot a^2 = \frac{\sqrt{3}}{4} a^2$$

$$A_1 = \left(1 - \frac{1}{4}\right) A_0 = \frac{3}{4} A_0$$

$$A_2 = \left(1 - \frac{1}{4}\right) A_1 = \left(\frac{3}{4}\right)^2 A_0$$



Iteration 2

We cut out one fourth of the area at each step.  
So, naturally area is reduced by  $\frac{3}{4}$  at each step.

# The Sierpinski Triangle: Area

Now let's find the area of triangle(s):

## Area Calculation

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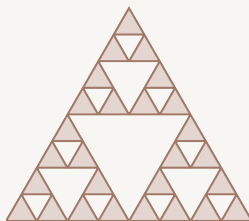
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...

$$A_n = \left(1 - \frac{1}{4}\right)^n A_0 = \left(\frac{3}{4}\right)^n A_0$$

$$\lim_{n \rightarrow \infty} A_n = 0$$



Iteration 3

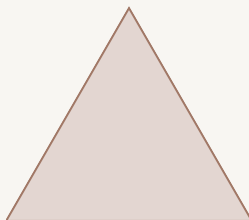
**The area of the Sierpinski Triangle is zero even though it is a 2D shape.**

# The Sierpinski Triangle: Perimeter

Now let's find the perimeter:

## Perimeter Calculation

$$P_0 = 3a$$



Iteration 0

# The Sierpinski Triangle: Perimeter

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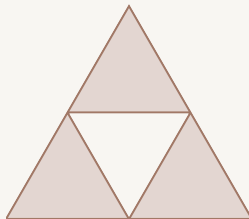
## Perimeter Calculation

$$P_0 = 3a$$

$$P_1 = P_0 + 3 \cdot \frac{a}{2} = \frac{9}{2}a = \frac{3}{2}P_0$$

We add half of the previous perimeter at each step. This is because we add the same number of triangles as the last step, but each triangle is half the size of the previous one.

As a result, the perimeter grows by a factor of  $\frac{3}{2}$  at each step.



Iteration 1

# The Sierpinski Triangle: Perimeter

Now let's find the perimeter:

## Perimeter Calculation

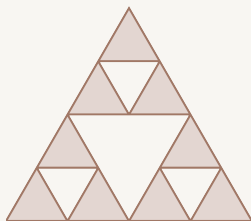
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Iteration 2

# The Sierpinski Triangle: Perimeter

Now let's find the perimeter:

## Perimeter Calculation

$$P_0 = 3a$$

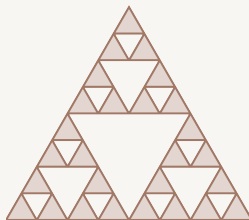
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...

$$P_n = \frac{3}{2}P_{n-1} = \left(\frac{3}{2}\right)^n P_0$$

$$\lim_{n \rightarrow \infty} P_n = \infty$$



Iteration 3

**The Sierpinski Triangle has infinite perimeter but zero area.**

We've now seen something strange, and precise:

- A **2D shape** (Sierpinski Triangle) with **zero area**.
- A shape built from a rule so simple a child could apply it by hand.
- No matter how far we zoom in, there's always more detail.

## Handoff to Speaker 2

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That number is exactly what Hausdorff was reaching for in 1918 — and it's where Speaker 2 picks up the story.

# Fractal Dimension

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*Let's talk about a property of fractals...*



# Fractal Dimension

What does it mean for a shape to have a **non-integer** dimension?

# Dimensions We Know

We are all familiar with integer dimensions:



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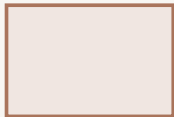
We are all familiar with integer dimensions:



Line



**1D**



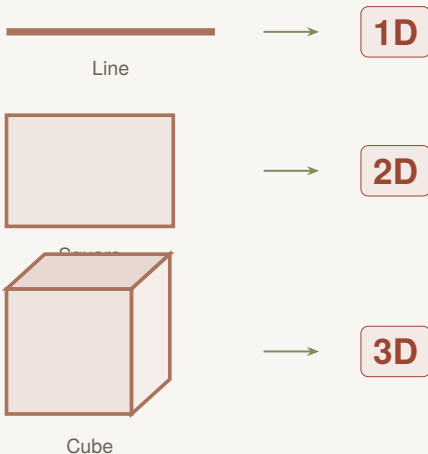
Square



**2D**

# Dimensions We Know

We are all familiar with integer dimensions:



*Dimensions are always whole numbers... right?*

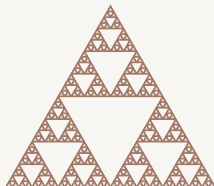
# What About This One?

What dimension does this  
shape have?

*Any guesses?*

- It lives in 2D space...
- But it has **zero area**
- And **infinite perimeter**

It's not quite 2D, but it's clearly more than  
1D...



Sierpinski Triangle

# What About This One?

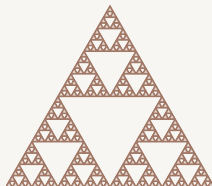
## Sierpinski Dimension

The Sierpinski Triangle has a fractal dimension of:

$$D \approx \mathbf{1.585}$$

!!

A **non-integer** dimension — between a line (1D) and a filled surface (2D).



$$D = \log 3 / \log 2 \approx 1.585$$

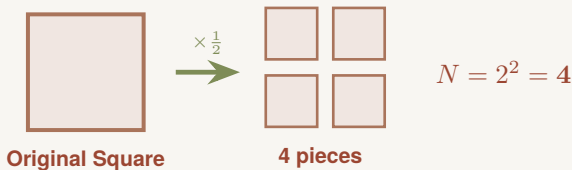
# Self-Similarity & Scaling

**Scaling a Line ( $D = 1$ ):** Scale by  $r = \frac{1}{2} \implies$  **2 pieces.** Scale again by  $\frac{1}{2} \implies 2 \times 2 =$  **4 pieces.**



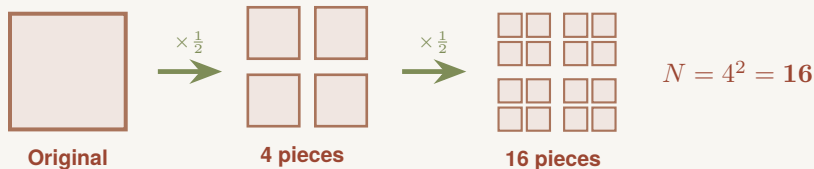
# Self-Similarity & Scaling

**Scaling a Square ( $D = 2$ ) — Step 1:** Scale by  $r = \frac{1}{2} \implies$  **4 self-similar pieces** ( $2 \times 2$  grid).



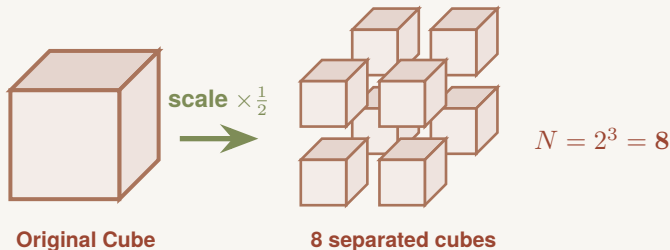
# Self-Similarity & Scaling

**Scaling a Square ( $D = 2$ ) — Step 2:** Scale each piece by  $\frac{1}{2}$  again  $\implies 4 \times 4 = 16$  self-similar pieces ( $4 \times 4$  grid)!



# Self-Similarity & Scaling

**Scaling a Cube ( $D = 3$ ):** Scale by  $r = \frac{1}{2} \implies$   
**8 self-similar pieces** ( $2 \times 2 \times 2$  grid in 3D).



# Self-Similarity & Scaling

## Do You See a Pattern?

Each scaling step multiplies pieces by  $(1/r)^D$ !

| Shape  | Scale | 1st Step  | 2nd Step     | $D$      |
|--------|-------|-----------|--------------|----------|
| Line   | $1/2$ | $2 = 2^1$ | $4 = 2^2$    | <b>1</b> |
| Square | $1/2$ | $4 = 2^2$ | $16 = 2^4$   | <b>2</b> |
| Cube   | $1/2$ | $8 = 2^3$ | $(64 = 2^6)$ | <b>3</b> |



$$D = ?$$

What if the shape is a **fractal**?

## The Pattern

$$\text{Number of pieces} = \left(\frac{1}{r}\right)^D$$

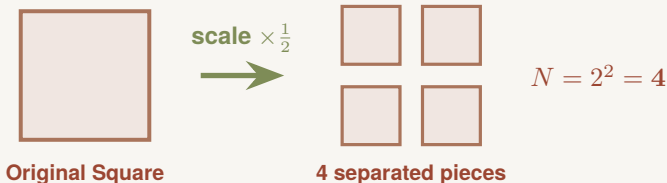
# The Self-Similarity Rule

After scaling by factor  $r$ , the number of self-similar pieces is:

$$N = \left( \frac{1}{r} \right)^D$$

**Square** ( $D = 2$ ) with scale factor  $r = \frac{1}{2}$ :

$$N = \left( \frac{1}{1/2} \right)^2 = 2^2 = 4 \text{ pieces}$$



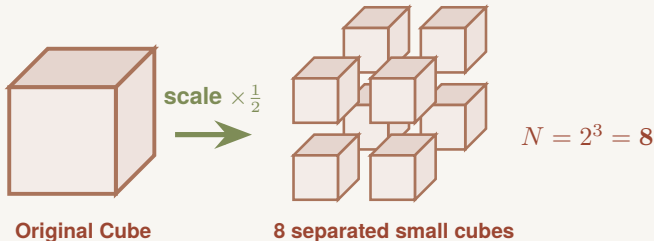
# The Self-Similarity Rule

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**Cube** ( $D = 3$ ) with scale factor  $r = \frac{1}{2}$ :

$$N = \left(\frac{1}{1/2}\right)^3 = 2^3 = 8 \text{ pieces}$$



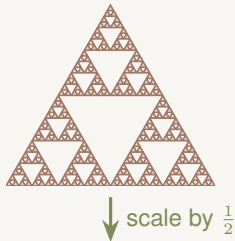
Can we use this same rule to find the dimension of a **fractal**?

# Dimension of the Sierpinski Triangle

Let's apply the self-similarity rule to the **Sierpinski Triangle**.

## Step 1

Scale the Sierpinski Triangle by  $\frac{1}{2}$ .

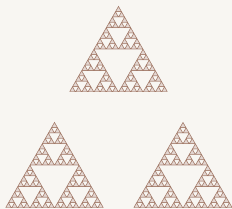


# Dimension of the Sierpinski Triangle

## Step 2 — Self-Similar Pieces

We get exactly **3** self-similar copies of the original triangle.

Each smaller triangle is a perfect replica, scaled by  $r = \frac{1}{2}$ .



**3 copies**

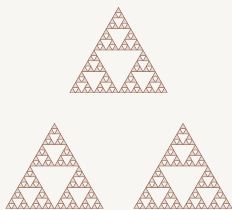
# Dimension of the Sierpinski Triangle

## Step 3 — The Equation

Using  $N = \left(\frac{1}{r}\right)^D$ :

$$3 = \left(\frac{1}{1/2}\right)^D$$

$$3 = 2^D$$



**3 copies**

# Dimension of the Sierpinski Triangle

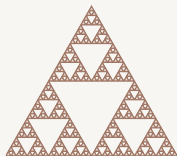
## Step 4 — Solving for $D$

$$2^D = 3$$

$$D \log 2 = \log 3$$

$$D = \frac{\log 3}{\log 2}$$

$$D \approx \mathbf{1.585}$$



$$D \approx 1.585$$

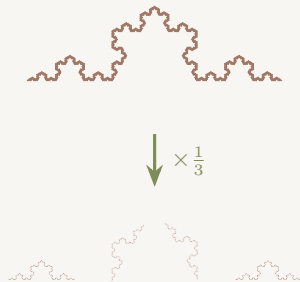
**Not an integer!** — strictly between 1D (line) and 2D (surface).

# Similarly: The Koch Curve

The **Koch Curve** can be divided into **4** self-similar pieces, each scaled by  $r = \frac{1}{3}$ .

## Setup

- Scale factor:  $r = \frac{1}{3}$
- Self-similar pieces:  $N = 4$



**4 copies**

## Similarly: The Koch Curve

### Koch Dimension

$$N = \left(\frac{1}{r}\right)^D$$

$$4 = 3^D$$

$$D = \frac{\log 4}{\log 3}$$

$$D \approx \mathbf{1.262}$$



$$D \approx 1.262$$

Also **non-integer** — between a line (1D) and a surface (2D), but closer to a line than the Sierpinski Triangle.

# The General Theory: Box-Counting Dimension

The **Self-Similarity Dimension** works beautifully for self-similar fractals. But what about more complex shapes?

## Box-Counting Dimension

The fractal dimension  $D$  is defined as:

$$D = \lim_{\varepsilon \rightarrow 0} \frac{\log N(\varepsilon)}{\log(1/\varepsilon)}$$

where  $N(\varepsilon)$  is the number of boxes of side length  $\varepsilon$  needed to cover the fractal.

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## Intuition

- Cover the fractal with a grid of boxes of size  $\varepsilon$
- Count how many boxes  $N(\varepsilon)$  the fractal touches
- Shrink  $\varepsilon \rightarrow 0$  and observe how  $N(\varepsilon)$  grows
- The **rate of growth** tells us the dimension

For a deeper dive, see [3Blue1Brown: Fractals are typically not self-similar](#)

## Handoff to Speaker 3

We've now uncovered a remarkable property of fractals:

- They have **non-integer dimensions** — genuinely “between” the familiar 1D, 2D, 3D.
- The Sierpinski Triangle lives in dimension  $D \approx 1.585$ .
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That's exactly where Speaker 3 takes us — fractals in nature, art, and computation.

# Fractals in the Real World

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*We know what a fractal is, and how to measure it...*



# Fractals in the Real World

Where does all of this actually **show up**?

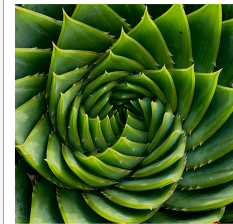
# Nature draws with one rule, repeated



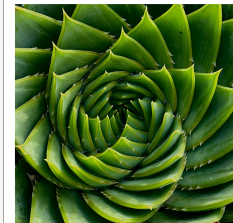
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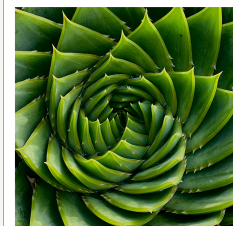
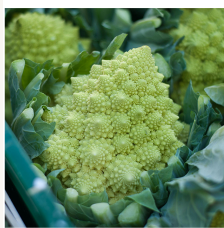
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Broccoli, ice, leaves, cones — the **same shape at every zoom**.

# A tree built from nothing but squares

**3 levels**



Pythagorean tree, after A. E. Bosman (1942), drawn at  $45^\circ$ .

# A tree built from nothing but squares

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7 levels



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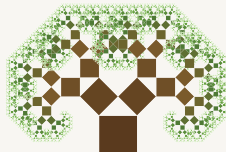
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Stand a **right triangle** on each square's top edge and grow a new square on each leg — so  $a^2 + b^2 = c^2$  sits at every single joint.

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Stand a **right triangle** on each square's top edge and grow a new square on each leg — so  $a^2 + b^2 = c^2$  sits at every single joint.

Nobody designed a single branch. **One rule**, repeated twelve times, and a tree appears.

Pythagorean tree, after A. E. Bosman (1942), drawn at  $45^\circ$ .

## tilted rule



## tilted rule



- At  $45^\circ$  the two branches match **exactly** — too perfect to be real

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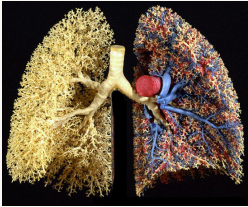
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**One number changed** — and a diagram turns into a plant.

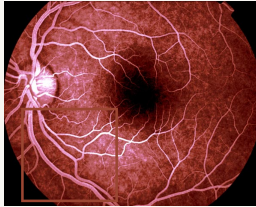
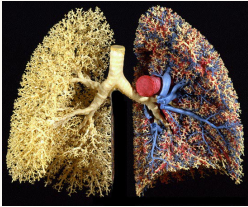
# Branching packs a huge surface into a small space



airways and blood vessels of the lung    vessels in the retina    the same vessels, zoomed in

Lung branching figures: Weibel (1963).

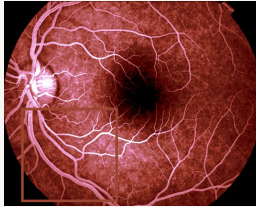
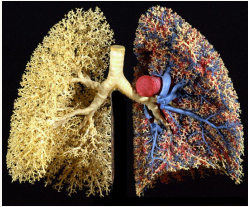
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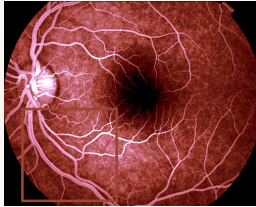
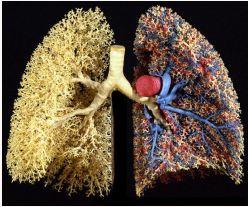
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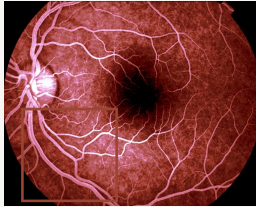
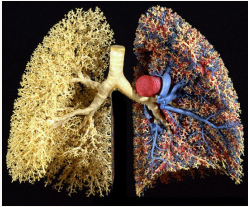


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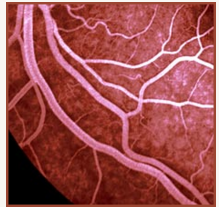
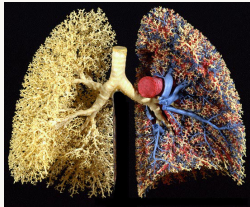


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Branching again and again is nature's cheapest way to get **area out of volume**.

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Rule 2 draws a **small copy of the whole plant**. That single line is where all the detail comes from.

# Fractal compression: a whole plant in 28 numbers

2,000 dots



Barnsley fern: an iterated function system (Barnsley, *Fractals Everywhere*, 1988), plotted by the chaos game.

# Fractal compression: a whole plant in 28 numbers

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- Four rules, each a **shrink-and-tilt** of the whole picture, applied at random

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Store the **rule**, not the pixels — the idea behind **fractal image compression**.

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# Game worlds are grown from a rule, not stored

**1 layer of bumps**



Terrain generated here by summing 7 layers of noise, each half the size of the last (fractal Brownian motion).

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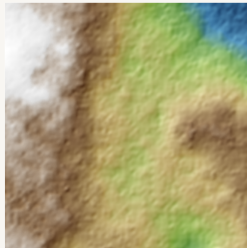
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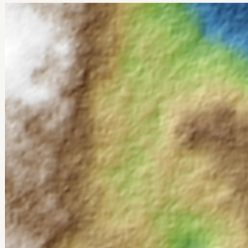
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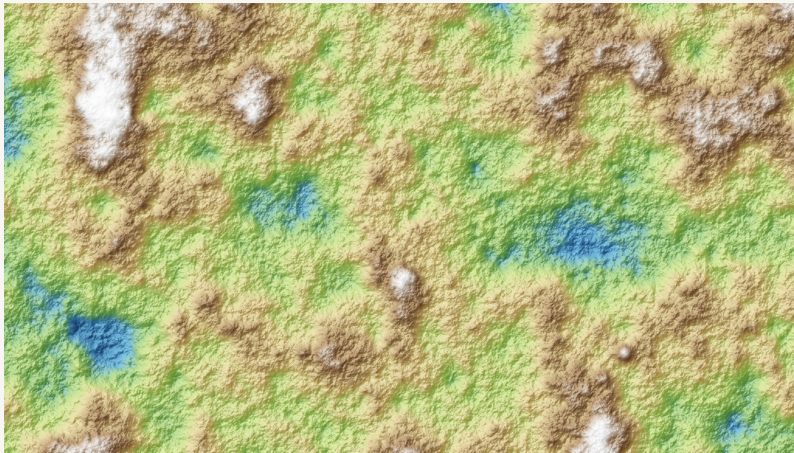
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Add **smaller bumps at every scale** — the Koch idea — and a handful of numbers becomes an endless mountain range. Store the rule, not the map.

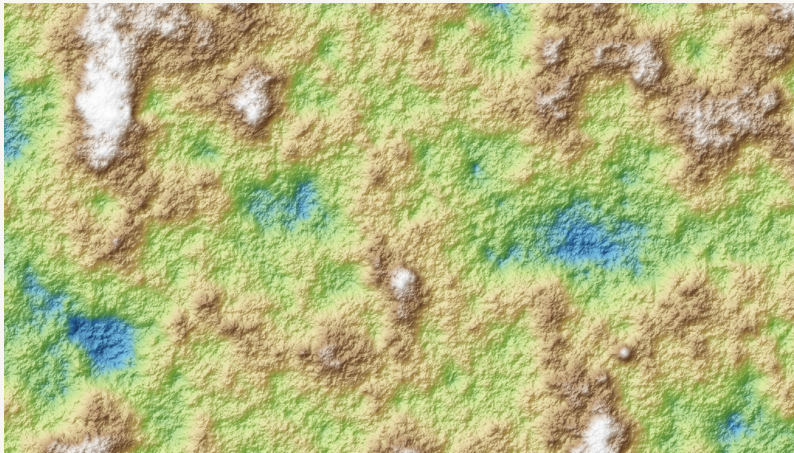
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# Keep stacking layers and you get a world



15 layers of value noise, each half the width and  $0.55 \times$  the height of the one before.

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Same rule, **15 layers** instead of 7 — ridges, valleys, snow lines and lakes. Not one of them was drawn by hand.

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A fractal is a **model** of nature — and every good model knows the range where it works.

# One rule, repeated, explains what smooth geometry couldn't

## Part 1 — the surprise



zero area,  
infinite edge

# One rule, repeated, explains what smooth geometry couldn't

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trees, lungs, ferns,  
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*Thank you — questions welcome.*

- B. Mandelbrot, “How long is the coast of Britain? Statistical self-similarity and fractional dimension,” *Science*, 1967.
- B. Mandelbrot, *The Fractal Geometry of Nature*, 1982.
- E. R. Weibel, *Morphometry of the Human Lung*, Springer, 1963.
- M. F. Barnsley, *Fractals Everywhere*, Academic Press, 1988 (iterated function systems, the fern).
- A. E. Bosman, *Het wondere onderzoekingsveld van de vlakke meetkunde*, 1957 (Pythagorean tree, drawn 1942).
- A. Fournier, D. Fussell, L. Carpenter, “Computer rendering of stochastic models,” *Comm. ACM*, 1982 (fractal terrain).
- 3Blue1Brown, “Fractals are typically not self-similar” (video).